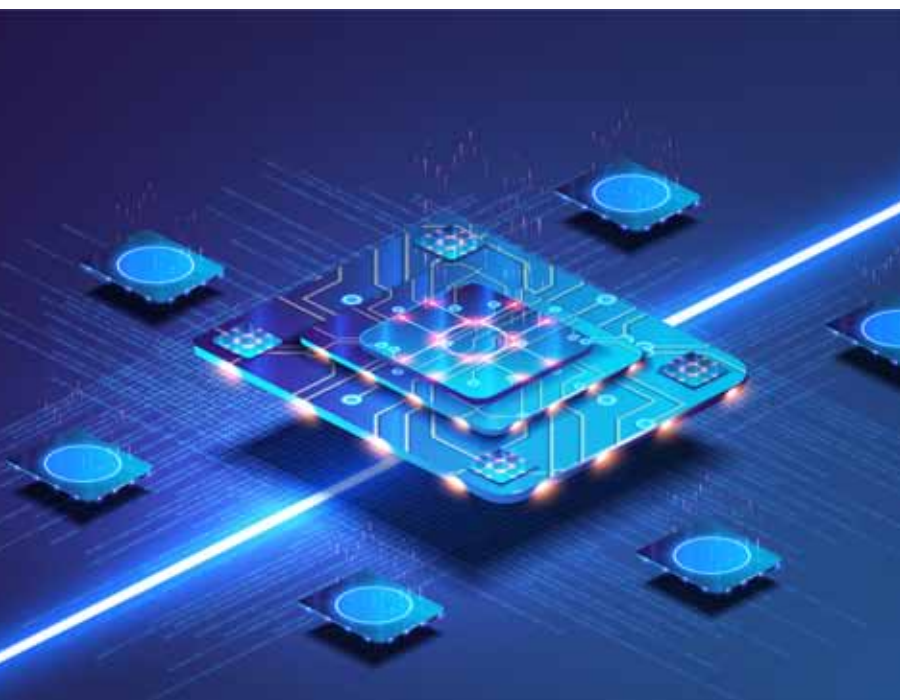




Quantum Technologies And Their Applications In Industry 4.0

The changes in the world each day are unlike anything humanity has ever encountered before in terms of size, breadth, and intricacy. With the revolution of Industry 4.0, how can quantum technologies find themselves to be a part of it? Chiranjeevi Aradhya, Senior Software Engineer at Collins Aerospace, Cedar Rapids, Iowa tells us how quantum technologies can be used



Chiranjeevi Aradhya

Quantum theory states that all energy and matter is made up of particles known as quanta. The long-term aim of quantum information technology research is to develop control and manipulation methods of quantum systems to extend information processing beyond the capabilities of the classical world. The usage of quantum concepts in engineering to solve difficult technicalities in computers, communications, sensing, chemistry, encryption, imaging, and mechanical issues will be quite common in the future. For example, the quantum computer can do things that even the most sophisticated supercomputers cannot.

Quantum and classical computers differ in three primary ways. First, quantum computing does not use bits that are either zero or one; it uses qubits that have the ability to act as either a zero or a one at the same time. Second, qubits reside in an entangled state rather than being separate entities. We can say, these two characteristics provide qubits the capacity to store exponentially more information than traditional computers. The third distinction is that a quantum computer operates in a fundamentally different manner. It analyses all possible outcomes simultaneously and utilises constructive interference to decide on a probable solution that is just right.